



# An investigation into the pyrolysis and oxidation of bio-oil from sugarcane bagasse: Kinetics and evolved gases using TGA-FTIR

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## ARTICLE INFO

Editor: Xianwei Liu

### Keywords:

Bio-oil  
Pyrolysis  
Oxidation: FTIR  
TGA  
Kinetics

## ABSTRACT

Bio-oil produced from the pyrolysis of sugarcane bagasse has the potential to be used as a sustainable and renewable energy source. In the present study, a non-isothermal thermo-gravimetric analysis (TGA) of the pyrolysis (in N<sub>2</sub> atmosphere) and combustion (in the air) of bio-oil from sugarcane bagasse was investigated at three heating rates: 5, 10, and 20 °C/min. The sample was heated from room temperature up to 900 °C and the evolved gases in the TG furnace were carried to a Fourier transform infrared (FTIR) cell where the composition of the gases and the functional groups present there were analyzed. A global kinetic analysis was performed to obtain the Arrhenius kinetic parameters for the pyrolysis and oxidation of the bio-oil using the distributed activation energy model. Three distinct stages, namely; low-temperature oxidation (LTO), fuel decomposition (FD), and high-temperature oxidation (HTO) were observed during the oxidation of bio-oil. The initial devolatilization of the oxygenated compounds observed during pyrolysis was similar to the LTO stage observed during combustion. The intensity of the CO<sub>2</sub> FTIR peaks seen during the bio-oil combustion was 10 times the intensity of the CO<sub>2</sub> peaks attained during pyrolysis. The TGA-FTIR analysis of the sugarcane bagasse bio-oil sheds new light on its thermal degradation/oxidation characteristics.

## 1. Introduction

Energy consumption in developing countries has doubled in the period between the years 2000 and 2014 [1] and by 2025 the worldwide energy demand is expected to grow by more than 50% [2]. Lignocellulosic biomass is now considered a renewable resource for the production of biofuels and several specialty chemicals. Biomass valorization to energy offers an ideal substitute to reduce and replace petroleum-based fuels as a renewable energy source. Lignin, which is one of the primary components of biomass is highly aromatic and several studies [3,4] have reported its potential in producing phenolic high-value bio-oils. Pyrolysis of biomass is a convenient method of transforming it into bio-oils that can be used as cheap fuels wholly or blended with fossil fuels to improve various combustion properties like octane number, cetane number, sooting propensity, etc. For example,

bio-ethanol is largely blended with gasoline (by up to 15%) as an octane booster. Oxygenate rich bio-oils [5] have also been blended as diesel fuel additives to reduce the formation of soot and particulate matter emissions in diesel engines. Various biomass resources like rice husk [6], kraft lignin [7], alkali lignin [8], wood waste [9], palm kernel [10], cottonseed [11], palm biomass [12], and others [13] have been used for the production of bio-oil through pyrolysis.

Sugarcane Bagasse (SCB), which is the fraction of biomass left after the juice is extracted from the sugarcane, is a cheap, promising, and renewable resource for the production of bio-oil. It is a fibrous residue made up of cellulose (50%), hemicellulose (25%), and lignin (25%) [14]. SCB is one of the main agricultural residues in Colombia and it is estimated that 25 M tons/year of SCB are produced and nearly 85% of it is used for steam generation in the sugar industry [15]. SCB has a high content of volatile material (80% w/w) which has a high potential to

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<https://doi.org/10.1016/j.jece.2021.106144>

Received 2 March 2021; Received in revised form 16 June 2021; Accepted 28 July 2021

Available online 31 July 2021

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convert into oxygen-rich fuels [16]. During pyrolysis, the cellulose and hemicellulose present in SCB undergo chemical transformation into oxygenated sugar-derived products such as furfural, monosaccharides, and linear oxygenated compounds as alcohols, aldehydes, and ketones that have shown very good characteristics as fuels [17], as well as into phenolic compounds that represent 10% of the bio-oil from lignin [18].

Pyrolysis technologies enable the production of biofuels from biomass, garbage, or organic waste [19–21]. SCB pyrolysis represents an alternative to cogeneration and allows an integration option for bio-refineries for the manufacture of fuels and high-value compounds. Most bio-oils obtained from pyrolysis are characterized as poor fuels due to several issues like low calorific value, high moisture and oxygen contents, poor miscibility with petroleum fuels, propensity to cause corrosion, chemical instability, acidity issues, etc. Characterization of bio-oils is an important step in understanding their physical and chemical properties, so appropriate upgrading techniques like fast pyrolysis, catalytic cracking, catalytic hydrogenation, emulsification, etc. can be implemented to improve their physical and thermochemical properties. Advanced characterization techniques like NMR spectroscopy [7,22–24], gas chromatography-mass spectroscopy (GC-MS) [6], Fourier transform ion cyclotron resonance (FT-ICR) mass spectrometry [25], etc. can shed light on the fuel molecular structure but cannot provide insight on the thermal stability or pollutant gases formed during combustion. Such information is useful for the design of combustors where the combustion efficiency of the bio-fuels can be maximized while minimizing the emissions and particulate matters. The most widely used technology to burn pyrolysis bio-oils is aspersation combustion [26–29], in this technology there are multiple physical and chemical phenomena that occur in very short time scales due to the size of the liquid droplets are below 1 mm [19,30], within this technology the convective effects of mass and heat transfer, caused by the turbulent flow regime, play an important role in the combustion of bio-oil [31]. It is precisely because of the simultaneous occurrence of phenomena that the information that can be collected with thermogravimetric analyzes becomes valuable. These results allow us to approach the global chemical oxidation kinetics by isolating convective and diffusive effects. Techniques like thermo-gravimetric analysis coupled to Fourier transform infrared spectroscopy (TGA-FTIR) have been widely employed to investigate the oxidation kinetics characteristics and pollutant emission mechanisms of several samples like heavy fuel oil [32], rice husk [6], alkali lignin [8], wood waste [9], etc. TGA-FTIR has several advantages like high accuracy and sensitivity, small sample requirements in the order of a few milligrams, real-time measurements, etc. It can help study mass loss characteristics, thermal degradation pathways, kinetic parameters, and on-line analysis of gases and functional groups evolved from the sample [33].

The analysis of TGA-DTA is a tool that allows the characterization of the evaporation of the fuel. Also, as has been done by other authors [25], it allows identifying the formation of unburned and related oxidation zones [34], although the analysis of TGA does not represents the combustion conditions in a spray burner if it offers the fundamental information on the dynamics of evaporation, and the chemical kinetics of pyrolysis and global oxidation.

The TGA-FTIR analysis allows not only to assess the evolution of mass during the heating of biofuel and to determine its kinetic parameters; it is also possible to identify the evolution of the functional groups released and permanent gases such as CO<sub>2</sub>, N<sub>2</sub>, CO, CH<sub>4</sub> during heating in real-time. There are few studies focused on the comparison of the pyrolysis and oxidation of bio-oil from biomass using TGA as is the case of bio-oil produced from rice husk [35], bio-oil from swine manure [36], as well as the aqueous fraction of wheat stalk bio-oil pyrolysis [37], in these works the authors agree that the oxidation of the bio-oil occurs in three stages: the evaporation of volatiles, the formation of high molecular weight volatiles, and the oxidation of char. Several models have been adjusted to calculate the kinetic parameters of oxidation and pyrolysis of the bio-oil; however, the complexity of the bio-oil composition

does not allow its results to be extrapolated and a better description of the oxidation is needed.

The objective of this work is to investigate the pyrolysis and oxidation of SCB bio-oil using TGA-FTIR. Thermal degradation and oxidation of bio-oil were analyzed using thermo-gravimetric (TG) and differential thermo-gravimetric (DTG) curves. Kinetic analysis of the bio-oil pyrolysis and oxidation was carried out to estimate the Arrhenius kinetic parameters using the distributed activation energy model by Miura and Maki [38,39]. From the evolution of the activation energy as a function of the conversion, the reaction stages, and the role that oxygen plays during oxidation are clarified. The composition of the evolved gases in real-time was analyzed using FTIR spectroscopy.

## 2. Materials and methods

### 2.1. Bio-oil production

Fresh SCB bio-oil was produced by pyrolysis at a temperature of 550 °C. It was produced in a spoon reactor using a horizontal electric furnace. A 30 cm long quartz tube with a 3 cm diameter was used and flushed with 50 ml/min of N<sub>2</sub> to provide an inert atmosphere. The carrier gas and the furnace were set at 550 °C and a stainless-steel spoon containing 1.5 g of SCB was introduced into the hot zone, reaching heating rates between 100 and 200 °C/s. The products from the pyrolytic reactions were immediately condensed in a U-type stainless-steel trap, submerged in an ice bath (see Fig. 1).

### 2.2. TGA-FTIR

TG analysis was performed on a Mettler Toledo TGA/DSC 1 (thermal gravimetric analysis/differential scanning calorimetry) instrument equipped with a GC 200 gas controller and an auto-sampler. For each test, approximately 10 mg of bio-oil was used and it was purged for 15 min at room temperature with a flow of 50 ml/min of N<sub>2</sub> (99.9999% purity) or ZERO air to displace CO<sub>2</sub> and water vapor introduced into the TG furnace during sample insertion. The samples were heated up to 900 °C at three heating rates of 5, 10, and 20 °C/min.

The gases evolved from the bio-oil during heating were transported to an IR Cell through a heated line (Thermo Scientific TGAFTIR Interface Nicolet iZ10) which was constantly maintained at 190 °C and equipped with a Swagelok 15 µm filter to remove any solid particles. The IR cell was kept at 200 °C to avoid condensation of vapors produced during thermal decomposition of the bio-oil. FTIR spectra of the gases were collected during the TGA heating ramp every 25 s with a resolution of 4 cm<sup>-1</sup> in the region of 400–4000 cm<sup>-1</sup> using a Thermo Scientific Nicolet iS10 connected to Nicolet iZ10 interface equipped with a deuterated triglycine sulfate (DTGS) detector and KBr beam splitter.

### 2.3. Kinetic modeling

The distributed activation energy model of Miura and Maki [38,39] was used in the present work to model the SCB bio-oil pyrolysis and oxidation. This model has been widely used by several researchers [32, 40,41] since it allows us to find both the kinetic parameters (activation energy, E<sub>a</sub>, and pre-exponential factor, A) and the ability to re-plot the TG profiles for comparison with experimental data. The Arrhenius based Miura and Maki model approximates the reaction rate using a first-order equation represented by Eq. 1.

$$\frac{d\alpha}{dT} = \frac{k_0}{\beta} \exp\left(-\frac{E_a}{RT}\right) f(\alpha) \quad (1)$$

Where  $\alpha$  is conversion defined by Eq. 2. and  $\beta$  is the heating rate.

$$\alpha = \frac{m - m_f}{m_i - m_f} \quad (2)$$

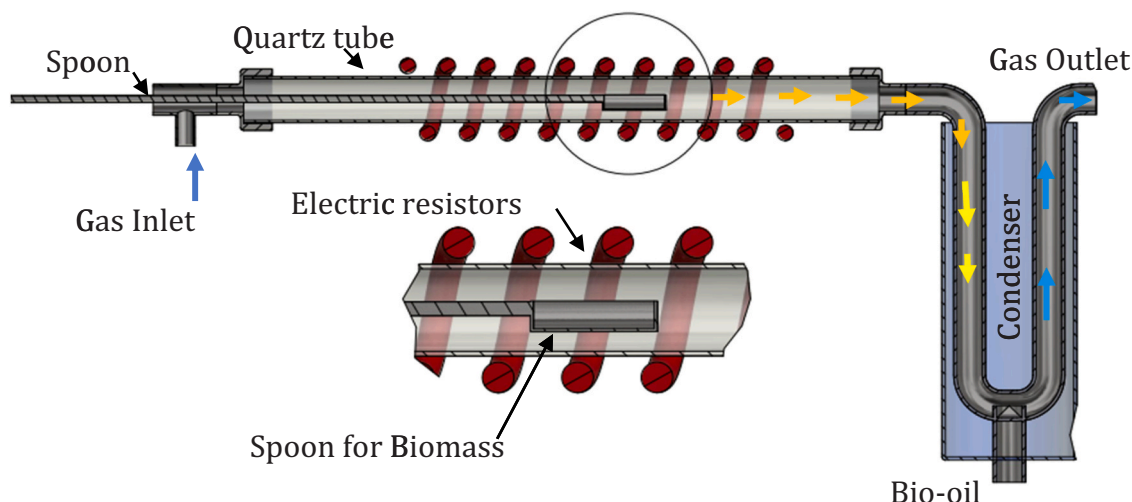


Fig. 1. Experimental set-up for SCB bio-oil production.

The terms  $m_i, m_f$  are the initial and final weights.  $E_a$  and  $k_0$  correspond to selected conversion values and these can be obtained from the slope and intercept of each Arrhenius plot using Eq. 3 for each of the three heating rates employed; 5, 10, and 20 °C/min.

$$\ln\left(\frac{d\alpha}{dT}\beta\right) = \ln[f_0][k_0 * f(\alpha)] + \left(-\frac{E_a}{RT}\right) \quad (3)$$

### 3. Results and discussion

The physical, chemical, and thermo-chemical properties of the SCB bio-oil produced in the present work are reported in Table 1. The physical characteristics of bio-oil are related to the lignocellulose composition of the biomass and the pyrolytic process conditions such as temperature, heating rate, and retention time of the vapors in the reactor [42,43]. SCB bio-oil has a high water content of 32.5 wt%, which hinders ignition and increases the Ignition Delay Time (IDT) due to the reduction in the rate of vaporization of the droplet in a burner [44]. The acidity of SCB bio-oil is of interest, due to the presence of oxygenated chemical compounds produced during pyrolysis and the organic acids in the bio-oil react with the storage materials and catalyze the formation of high molecular weight compounds during heating. The Total Acid

Number (TAN) of SCB bio-oil measured was 95 mg KOH/g of bio-oil which is close to the typically expected value of 100. The residual solids content of the pyrolysis bio-oil, measured by filtration after dilution with methanol, is less than the prescribed tolerance value of 2.5 wt%. The Lower Heating Value (LHV) is close to 15 MJ/kg, which makes the bio-oil a good quality fuel, according to the ASTM D 7544 burner fuel standard for fast pyrolysis bio-oils. These values indicate that the SCB bio-oil has good properties needed for a bio-fuel on the market and to promote its acceptance as a fuel [45].

The elemental (CHNSO) values of SCB bio-oil are also within the range of values observed from various pyrolysis bio-oils [30,43,46,47].

#### 3.1. Pyrolysis of SCB bio-oil

The TG and DTG results of the pyrolysis (in N<sub>2</sub> atmosphere) of SCB bio-oil using the three heating rates (5, 10, and 20 °C/min) are shown in Fig. 2. The bio-oil undergoes thermal decomposition due to the absence of oxygen and a notable mass loss is observed between 70 and 200 °C. This region is characteristic of the elimination of water - which forms a major portion of the bio-oil - and removal of acids, alcohols, and other aromatic oxygenate such as mono-phenols and furans [48]. DTG peaks are observed in this region at temperatures of 77, 107, and 119 °C at the heating rates of 5, 10, and 20 °C, respectively. The peaks move towards

Table 1  
Physical and chemical properties of SCB.

| Property                       | Units             | Value      | Method                      |
|--------------------------------|-------------------|------------|-----------------------------|
| Density (at 288 K)             | kg/m <sup>3</sup> | 1170       | ASTM D4052-11               |
| Kinematic viscosity (at 288 K) | cSt               | 8.14 ± 0.2 | ASTM D445-12                |
| Water (Karl fisher titration)  | wt%               | 32.5 ± 2   | ASTM E203                   |
| Total Phenolic content         | mg GAE/g bio-oil  | 3145 ± 35  | Folin-Ciocalteu colorimetry |
| Total acid number (TAN)        | mg KOH/g          | 95         | ASTM D664                   |
| Pyrolysis solids content, max  | wt%               | 1.5        | ASTM D7579                  |
| LHV                            | MJ/kg (wet basis) | 13.5       |                             |
| Ultimate analysis              |                   |            |                             |
| carbon                         | wt%               | 47.6       | EPA 440                     |
| hydrogen                       | wt%               | 3.3        | EPA 440                     |
| oxygen                         | wt%               | 15.6       | EPA 440                     |
| nitrogen                       | wt%               | 0.14       | EPA 440                     |
| sulfur                         | wt%               | 0.09       | ASTM D4294-10               |

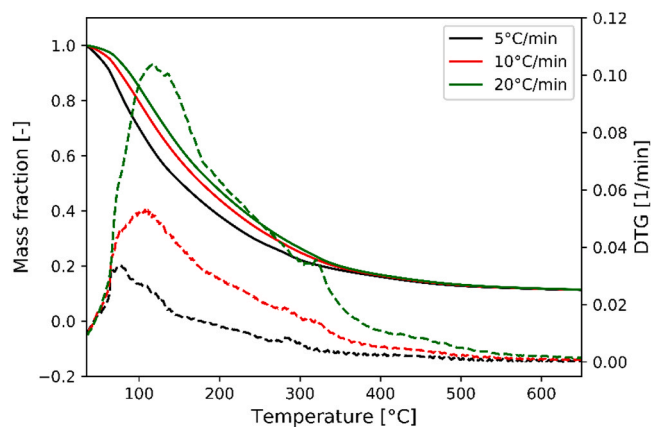


Fig. 2. TGA (continuous line) and DTG (dotted line) characteristics of SCB bio-oil pyrolysis.

higher temperatures when the heating rate is increased due to thermal hysteresis, which increases at higher heating rates. At temperatures between 200 and 350 °C, the decomposition of mono-sugars such as levoglucosan and poly-sugars such as cellobiose takes place as reported elsewhere [48,49]. The remaining bio-oil that underwent dehydration and crosslinking reactions [50–52] now forms carbonaceous structures that correspond to about 11.5% of the initial mass and are a result of the decomposition of pyro-lignin, which is the heaviest fraction of the bio-oil [53]. The rate of mass loss continues to fall steadily beyond 350 °C as the volatile content is depleted and the residue left at the end is mostly carbonaceous char that does not decompose further.

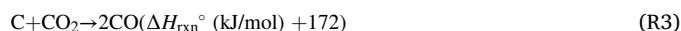
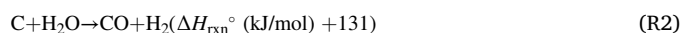
### 3.2. Oxidation of bio-oil

SCB bio-oil is a complex mixture of organic compounds, and a large number of parallel oxidation reactions occur due to the presence of oxygen in the bio-oil and the oxidizer air with an increase in temperature. Fig. 3 shows the TG and DTG histories of the bio-oil obtained during its oxidation (in the air) at the three heating rates. Analyzing the curves obtained, three zones are identifiable: low-temperature oxidation (LTO), fuel decomposition (FD), and high-temperature oxidation (HTO). During the LTO phase, the fuel undergoes processes like boiling and devolatilization, where low molecular weight compounds are eliminated. Aromatic oxygenated compounds have also been shown to oxidize at temperatures associated with the homogenous LTO phase [54–56]. The oxidation characteristics of the fuel at the LTO stage resemble closely with those observed during the initial stages of pyrolysis. Maximum weight loss is observed at temperatures of 91, 103, and 129 °C at heating rates of 5, 10, and 20 °C, respectively.

The fuel decomposition stage (FD) is characterized by a reduction in the rate of mass loss. This deceleration occurs mainly due to the decomposition of the fuel (thermal cracking) that requires the absorption of oxygen for the formation of intermediaries. In this stage, there are a large number of high molecular weight components in the plastic layer [57] formed by the carbonization and crosslinking reactions, similar to the behavior reported in the pyrolysis of biomass [42,58]. During this stage, the temperatures are sufficiently high for the mineral content present in the bio-oil such as potassium to promote catalytic decomposition reactions [59,60]. This thermal decomposition stage occurs in the range of 289–407 °C at 5 °C/min, 307–429 °C at 10 °C/min, and 339–449 °C at 20 °C/min. The low rate of thermal decomposition observed here is largely due to the stability of lignin, which is generally the main component remaining at these temperatures [53].

After the fuel decomposition stage, the solids formed pass to a stage in which high temperature oxidation reactions (HTO) occur as the activation energy necessary for the oxidation reactions in this stage is reached, the reactions R1 to R3 are very important because they are

responsible for the formation of carbon monoxide, carbon dioxide and hydrogen. The gasification reactions R2 and R3 are endothermic, however their products are highly combustible in the gas phase and this maintains the thermal balance [61].



The HTO stage is observed in the temperature ranges between 415 and 518 °C for a heating rate of 5 °C/min, 433–545 °C at 10 °C/min and 449–593 °C at 20 °C/min. The solids formed in the fuel decomposition stage (FD) now go through a thermal cracking process on their surface that allows the migration of oxygen by diffusion through its pores, resulting in an almost complete oxidation close to 99.6% of the total mass of bio-oil.

In the biooil spray combustion experiments, two stages are identifiable, the first the volatile combustion and the second the heterogeneous combustion of charred [34], although the experimental conditions of the two research systems are very different, the results of the TGA analysis allows to identify these stages and clarify how they occur.

Finally, Table 2 summarizes the possible zones found, the temperature at which the peaks identified by DTG and the weight loss rate at the peak appear. Something remarkable in these results is that as the heating rate of the peaks moves towards higher temperatures, this as suggested by other authors [62], may be related to the thermal delay that occurs at high heating rates [63], and therefore the increase in time during which the sample is exposed to a heating stage.

Fig. 4 shows the DTG curves of the bio-oil pyrolysis and oxidation at the three heating rates. The shape of the pyrolysis and oxidation curves during the LTO stage is similar until it reaches a temperature at which hysteresis is generated, which is characteristic of oxygen absorption [64].

It would be expected that the DTG analysis between 100 and 300 °C for the case of pyrolysis and oxidation at 20 °C/min would be very similar, however, it is evident that there are significant differences mainly due to two phenomena; The first stage of volatile release is difficult to replicate due to the experimental difficulties due to the evaporation of light components before carrying out the TGA analysis; and the second is the influence of the atmosphere, the oxygen present in the atmosphere could be adsorbed to low temperatures on the liquid surface and this phenomenon reduces the rate of evaporation weight loss. This difference between the behavior of pyrolysis and oxidation becomes less evident at a lower rate of heating because the exposure time at a given temperature increases, so the phenomenon is not appreciable.

### 3.3. Kinetic analysis of bio-oil oxidation

Kinetic modeling of the pyrolysis and combustion of the SCB bio-oil can help us understand the degradation/oxidation characteristics And it allows to characterize the phenomena associated with evaporation, fuel degradation, formation and oxidation of unburned, as well as knowing the mechanisms by which polluting gases are released. oxidation of SCB bio-oil is a multi-component oxidation process, and it is known that activation energy changes with the degree of conversion. The Miura and Maki model can be used to obtain a distribution of activation energies as a function of the reaction conversion ( $\alpha$ ) as shown in Fig. 5. In the plot, the colored area next to the curve is the confidence interval of the fitting lines for each  $\alpha$  selected.

In Fig. 5 it can be observed, in the case of oxidation and pyrolysis, that the activation energy increases gradually from  $\alpha = 0.1$ –0.8. This increase may be the result of the decomposition of high molecular weight compounds, which are the product of the dehydration reactions of the lignin oligomers present in bio-oil. The carbon formed is thermally

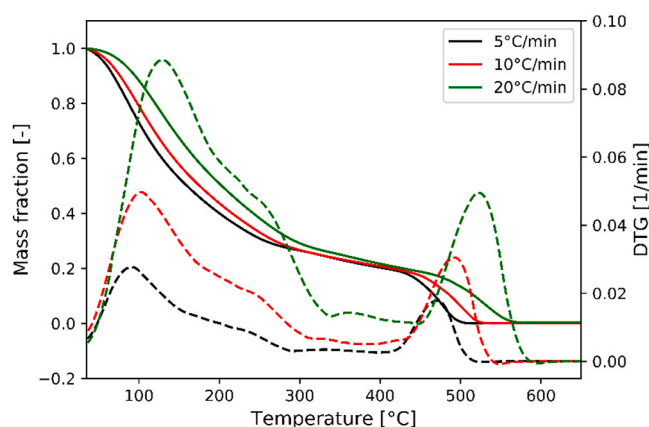
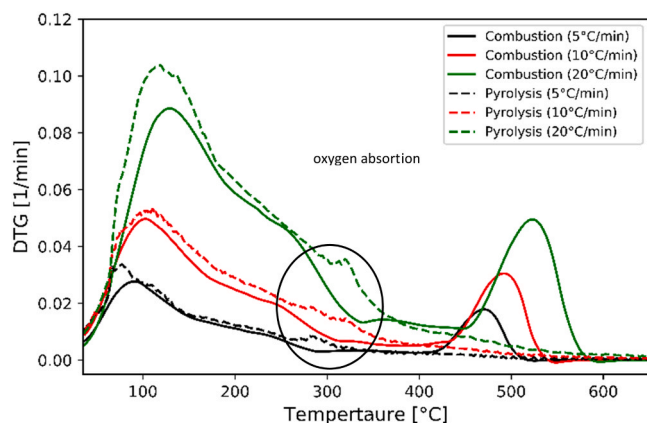
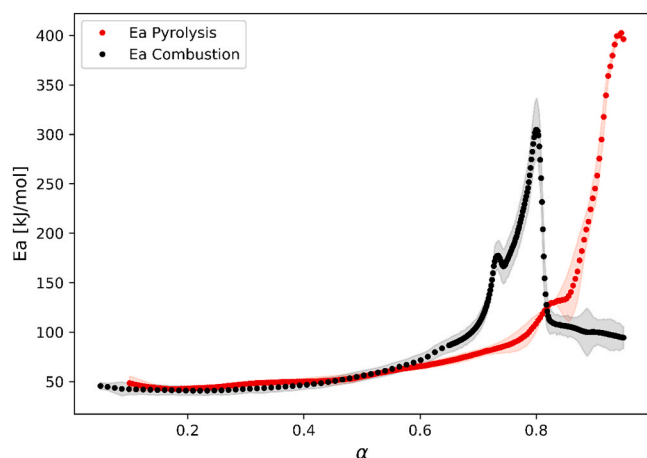


Fig. 3. TGA (continuous line) and DTG (dotted line) characteristics of SCB bio-oil oxidation.



**Table 2**Thermal degradation characteristics of PKS under nitrogen and air atmosphere with different heating rates of 5, 10, and 20 °C min<sup>-1</sup>.

| $\beta$ °C/min | Temperature range in zone °C |         |          |         | Peak temperature °C |     |     |     | Weight loss rate at the peak 1/min |       |       |       |
|----------------|------------------------------|---------|----------|---------|---------------------|-----|-----|-----|------------------------------------|-------|-------|-------|
|                | Zone I                       | Zone II | Zone III | Zone IV | Tp1                 | Tp2 | Tp3 | Tp4 | WLRp1                              | WLRp2 | WLRp3 | WLRp4 |
| N2-5           | 61–182                       | /       | 260–319  | /       | 78                  | /   | 287 | /   | 0033                               | /     | 0008  | /     |
| N2-10          | 61–185                       | /       | 266–325  | /       | 106                 | /   | 318 | /   | 0053                               | /     | 0014  | /     |
| N2-20          | 61–187                       | /       | 300–352  | /       | 118                 | /   | 319 | /   | 0103                               | /     | 0035  | /     |
| Air-5          | 55–171                       | 171–296 | 296–417  | 417–521 | 91                  | 235 | 317 | 469 | 0027                               | 0009  | 0003  | 0017  |
| Air-10         | 55–224                       | 224–302 | 302–425  | 425–548 | 102                 | 250 | 336 | 494 | 0050                               | 0020  | 0007  | 0030  |
| Air-20         | 55–190                       | 190–322 | 322–448  | 448–598 | 130                 | 261 | 363 | 523 | 0088                               | 0044  | 0014  | 0049  |

**Fig. 4.** DTG comparison of SCB bio-oil pyrolysis (dotted line) and oxidation (continuous line).**Fig. 5.** Activation energy vs. reaction conversion ( $\alpha$ ).

stable and requires more energy to achieve oxidation. However, in the case of oxidation, after reaching a maximum value at  $\alpha = 0.8$ , the calculated activation energy falls. This can be explained because when the maximum rate of reaction and heat release is reached, this released heat spontaneously promotes the oxidation of the structures that have not yet oxidized, generating a reduction in the activation energy calculated with this model.

During pyrolysis, for  $\alpha < 0.5$ ,  $E_a$  close to 65 kJ/mol was obtained. This value is close to the  $E_a$  values reported by other authors in the devolatilization stage of low molecular weight compounds [55]. At higher conversions,  $E_a$  reaches values close to 150 kJ/mol corresponding to the values associated with the pyrolysis reaction of lignin oligomers [65] found in the bio-oil. At this stage cross-linking chain reactions occur as a result of decay reactions or exposed functional groups in the carbonaceous solids formed [66].

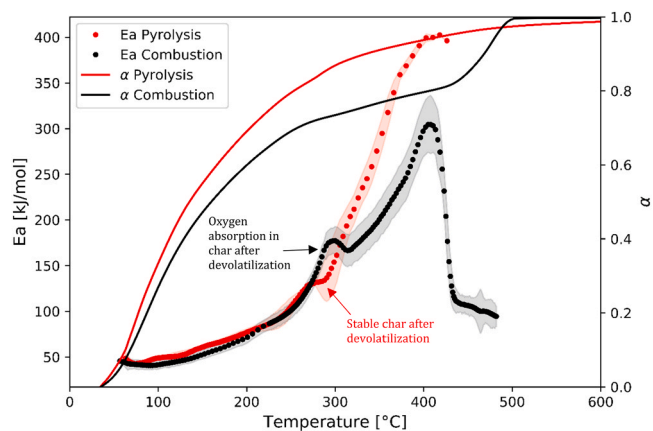
During oxidation of SCB bio-oil, it can be seen that the activation energy distribution shows two clearly defined peaks; the first at  $\alpha = 0.73$  which is just before the decomposition reactions of the fuel are triggered, the second peak seen at  $\alpha = 0.8$  corresponds to the HTO stage. The maximum  $E_a$  value in the FD stage was 179.4 kJ/mol and that during the HTO stage it was 303.47 kJ/mol.

Fig. 6 shows the plot of  $E_a$  versus temperature for both pyrolysis and oxidation at a heating rate of 5 °C/min. It can be seen that in the case of pyrolysis the temperature required to reach  $\alpha = 0.85$  is only 326 °C, whereas the same conversion is achieved at a higher temperature of 450 °C during oxidation. This is due to the oxidation of the char formed at the end of the FD stage and char oxidation requiring higher temperatures than devolatilization.

In Fig. 6 it can be seen that at temperatures lesser than 250 °C the activation energy values are similar for both pyrolysis and oxidation. This is mainly because similar processes like devolatilization occur during both processes. At temperatures between 250 and 320 °C, significant differences in the activation energies are observed between pyrolysis and oxidation, which is due to the absorption of oxygen before the FD stage as explained before. The  $E_a$  increases due to the absorption of oxygen by the carbonaceous solids formed. Significant differences in the  $E_a$  are also observed between pyrolysis and oxidation at temperatures greater than 320 °C. The apparent difference between the  $E_a$  values in the three oxidation stages (LTO, FD, and HTO) of SCB bio-oil shows the usefulness of the Miura and Maki model in capturing the reaction kinetics.

### 3.4. FTIR results

Coupling the FTIR cell with the TG furnace allows measuring the composition of the gases released during the pyrolysis and oxidation of the bio-oil using characteristic IR absorbance bands. The measured IR absorbance corresponds to the amount of energy absorbed by the vibrational modes of various bonds and functional groups present in the gases. This allows identifying and quantifying the composition of gases

**Fig. 6.** Activation energy vs. temperature until reaching  $\alpha = 0.85$  in each process.

such as carbon dioxide, carbon monoxide, and water, among others [67]. Furthermore, it is also possible to give a qualitative description of the presence of aromatic C—C, aliphatic C—H, and C—O bonds in the gases.

### 3.4.1. Evolved gases during pyrolysis

Fig. 7 shows the FTIR spectra of gases evolved during the pyrolysis of bio-oil at a heating rate of 5 °C/min. The main signals that produce a strong IR intensity are the stretching of the O—H group ( $3300\text{--}4000\text{ cm}^{-1}$ ) [10,48,68], unsaturated and saturated C—H stretch in aromatic and aliphatic compounds ( $2600\text{--}3100\text{ cm}^{-1}$ ), respectively, carbon dioxide ( $2360\text{ cm}^{-1}$ ), oxygenated functional groups such as aldehydes, ketones, and carboxylic acids C=O ( $1630\text{--}1780\text{ cm}^{-1}$ ), aromatic C=C bond ( $1500\text{--}1600\text{ cm}^{-1}$ ), and the bending vibrations in the fingerprint region (below  $1500\text{ cm}^{-1}$ ) [10,67,68].

As observed during the pyrolysis of bio-oil at a temperature of 200 °C (see Fig. 2), nearly 60% of the mass was lost due to devolatilization, including water, oxygenates, and volatile organic compounds [48,55]. Five signals are of interest and their evolution along with the temperature is shown in Fig. 8. The signals at  $1745\text{ cm}^{-1}$  and  $1840\text{ cm}^{-1}$  correspond to C=O vibrational stretching of aldehydes, ketones, and carboxylic acids (likely associated with acetic acid and acetol),  $1080\text{ cm}^{-1}$  correspond to C—O bond stretch related to the presence of oxygenated compounds (associated with the presence of anhydrides),  $3870\text{ cm}^{-1}$  for  $\text{H}_2\text{O}$ , and  $2360\text{ cm}^{-1}$  for  $\text{CO}_2$ .

The signal associated with water shows a maximum at a temperature close to 90 °C and corresponds to the evaporation of water present in the bio-oil. The functional groups related to oxygenated compounds (C=O and CO) show a peak in the range from 100° to 200 °C and the signal at  $1745\text{ cm}^{-1}$  is the most intense since aldehyde, ketone and carboxylic acid groups are abundant in the bio-oil. On the other hand,  $\text{CO}_2$  shows a considerable presence beginning at 70 °C with a maximum concentration at temperatures close to 300 °C. The evolution of  $\text{CO}_2$  completely ceases close to 500 °C, which indicates that it is a product of the decomposition of organic components in bio-oil.

### 3.4.2. Evolved gases during oxidation

Fig. 9 shows the FTIR spectra of the gases evolved during the bio-oil oxidation at a heating rate of 5 °C/min. In the region between  $1850\text{ and }500\text{ cm}^{-1}$  and temperatures below 200 °C, a pattern very similar to that obtained during bio-oil pyrolysis is observed, however, the intensity of the signal associated with  $\text{CO}_2$  is ten times higher. This is due to the gas phase oxidation of the evolved gases from the bio-oil during the LTO stage. The peak corresponding to  $\text{CO}_2$  which appears at  $2360\text{ cm}^{-1}$  and the confirmation peak at  $670\text{ cm}^{-1}$  in the fingerprint region is shown in Fig. 10.

As can be seen in Fig. 9, the carbon dioxide peak fits very well into the high-temperature oxidation region. This explains that the increase in mass loss measured in TG is because in this temperature range the

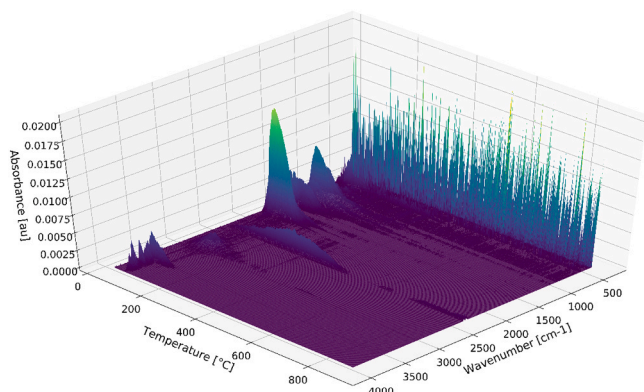


Fig. 7. FTIR spectra of SCB bio-oil pyrolysis in  $\text{N}_2$  at a heating rate of 5 °C/min.

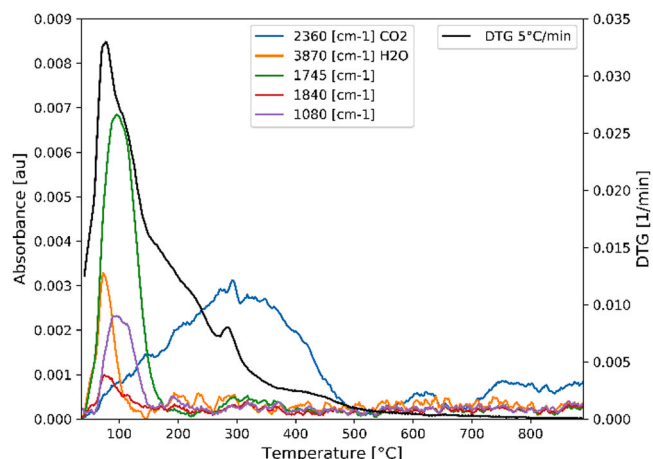


Fig. 8. Selected wavenumbers evolution with temperature during pyrolysis of SCB bio-oil at heating rate of 5 °C/min.

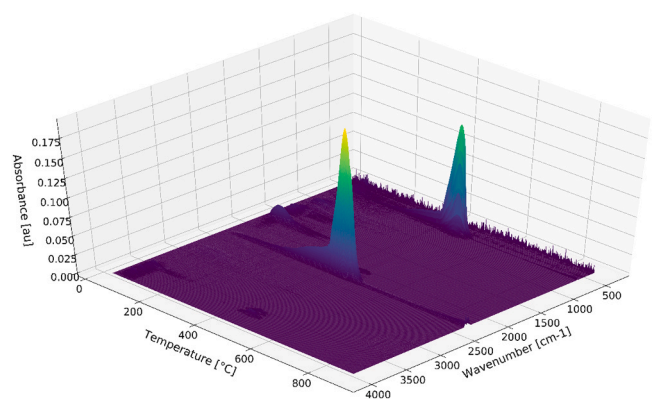


Fig. 9. FTIR spectra of oxidation of SCB bio-oil in the air at a heating rate of 5 °C/min.

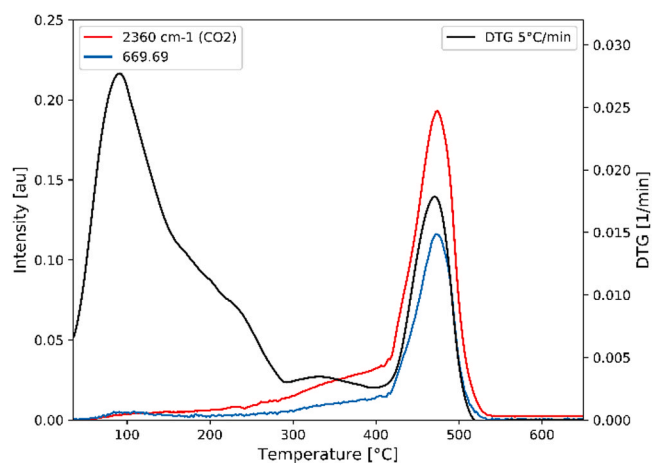


Fig. 10. FTIR intensity evolution for  $2360\text{ cm}^{-1}$ , and  $670\text{ cm}^{-1}$  wavenumbers.

greatest amount of oxidation gases such as  $\text{CO}_2$ , CO, and  $\text{H}_2\text{O}$  are released.

Fig. 11 shows the evolution of  $\text{H}_2\text{O}$  ( $3733\text{ cm}^{-1}$ ), CO ( $2119\text{ cm}^{-1}$ ), and NO ( $1762\text{ cm}^{-1}$ ) during the bio-oil oxidation at a heating rate of 5 °C/min. A prominent water peak appears at 73 °C due to the evaporation of water in the bio-oil and then it falls rapidly, disappearing at 140 °C. The water peak re-appears in the range of (300–400 °C) that

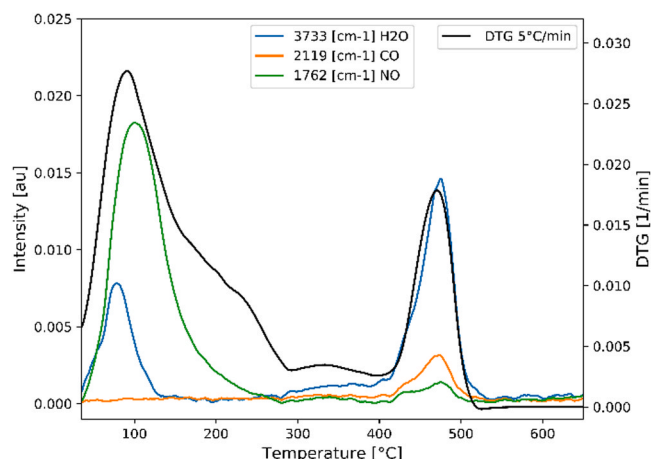


Fig. 11. FTIR intensity evolution for 3733, 2119, and 1762  $\text{cm}^{-1}$  during oxidation.

corresponds to the fuel decomposition stage (FD) where the absorption of oxygen at the surface occurs along with carbonization reactions [69, 70]. Between 400 and 500  $^{\circ}\text{C}$ ,  $\text{H}_2\text{O}$  evolution increases, and peak intensity can be observed because of oxidation at high temperatures. In this range, the formation of  $\text{CO}$  is also observed due to the partial oxidation of the macromolecules formed during the FD stage [71].

The nitrogen content of the bio-oil (0.14 wt%) is responsible for the fuel  $\text{NO}_x$  emissions during oxidation. The contribution due to thermal  $\text{NO}_x$  may or may not be relevant due to the absence of high temperatures ( $> 1500^{\circ}\text{C}$ ) needed for its formation. A characteristic signal of the  $\text{N}-\text{O}$  interaction is associated with the signal of ( $1762\text{ cm}^{-1}$ ) [12,72] and the  $\text{NO}$  signal attains a peak at  $105^{\circ}\text{C}$ , soon after the maximum devolatilization peaks at  $91^{\circ}\text{C}$ . This can be attributed to the homogenous oxidation of  $\text{N}$ -containing low molecular weight compounds released during the LTO stage. During the HTO stage, the  $\text{N}-\text{O}$  signal

corresponds to the nitrogen oxides formed due to the oxidation, of the carbonaceous macromolecules containing nitrogen atoms in their core structure.

Comparing the gases released during oxidation and pyrolysis, it is important to analyze the behavior of three gases, especially Methane, the functional group  $-\text{OH}$  - which can come mainly from water - and carbon dioxide. Fig. 12 shows the curves of the three gases selected using different heating rates, as can be seen both in pyrolysis and in oxidation, when increasing the heating rate the intensity of the signal peak increases and there is also a small displacement of the peak, this occurs because the gases released at a higher heating rate occupy the same volume of the measurement cell in a shorter time, so the relative concentration of the gases increases and this increases the absorbance, and the displacement of the peak is the effect of the thermal delay suffered by the sample at a higher heating rate.

The methane released in both processes, pyrolysis and oxidation, has been previously reported [73,74] as the result of the decomposition of the methoxy group ( $-\text{OCH}_3$ ) or the methyl ( $-\text{CH}_3$ ) or methylene group ( $-\text{CH}_2-$ ). The methoxy group is usually associated with the decomposition of lignin or its derivative compounds during pyrolysis and methyl or Methylene is associated with side chains in glycosidic compounds that come from the hemicellulose and lignin of biomass with which the bio-oil was produced. Fig. 12a and d shows the evolution of methane during pyrolysis and oxidation, respectively, the first peak may come from the decomposition of the methoxy groups from the pyrolytic lignin present in the bio-oil, and at high temperatures, it is produced by the effect of the decomposition of the side chains of the high molecular weight char formed. Under oxidation conditions, the intensity of the low signal is due to the formation of other gases such as water and carbon monoxide that dilute the gas, as reported in the case of the biomass by Ma et al. [62].

Water, on the other hand, can appear by evaporation of the water present in the bio-oil, by the formation of water during carbonization by dehydration reactions, and in air atmospheres as a result of the oxidation of the char. As can be seen in Fig. 12b, there is a defined peak at low

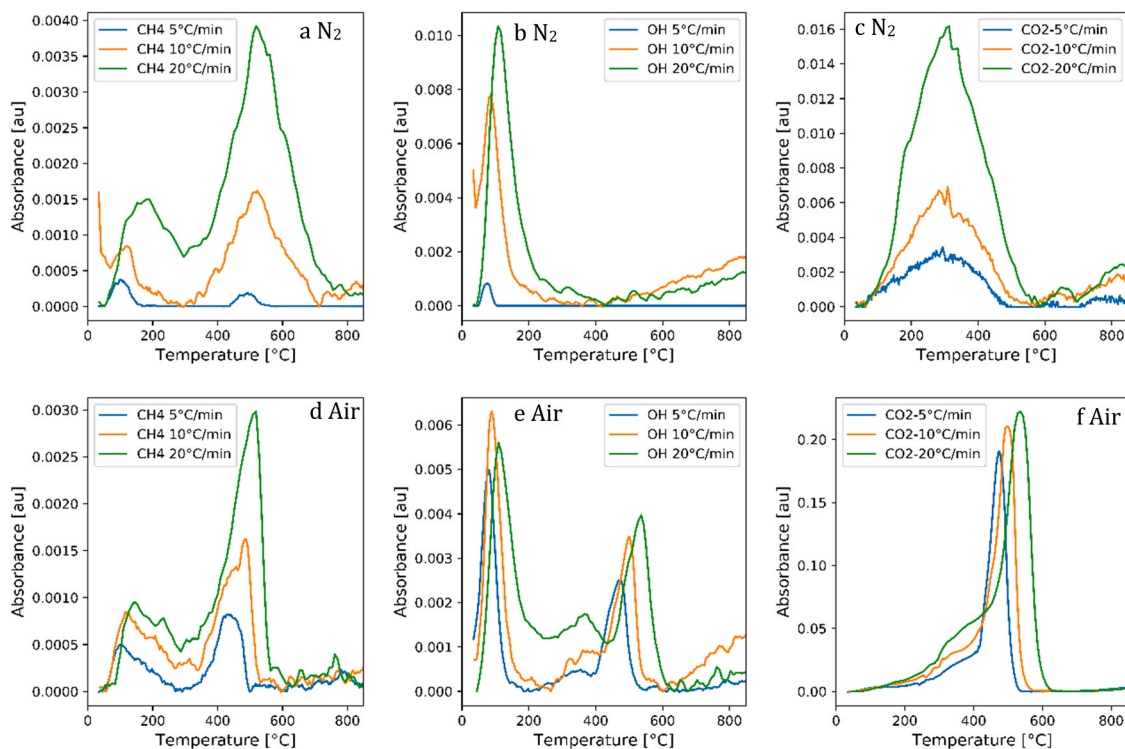


Fig. 12. evolution of gases released during pyrolysis (a, b, c) and during oxidation (e, f, g) the signals were assigned to the species methane ( $2850\text{ cm}^{-1}$ ),  $\text{CO}_2$  ( $2360\text{ cm}^{-1}$ ) and  $\text{OH}$  ( $3733\text{ cm}^{-1}$ ).



temperatures as a result of the evaporation of water present in the bio-oil, and above 400 °C a signal of considerable intensity begins to appear as a result of the condensation reactions responsible for the formation of char. On the other hand, in the case of Fig. 12e, there is a much-defined peak between 400 and 600 °C, which is the product of oxidation.

Carbon dioxide in pyrolysis atmospheres (Fig. 12c) can appear as a result of decarboxylation or decarbonylation as a route for thermal decomposition, for the case of biomass pyrolysis there are studies of TGA FTIR of hemicellulose [74,75] and cellulose [76,77] where two peaks of interest appear; the first associated with hemicellulose by decarboxylation reactions of oxygenated groups ( $\text{—COOH}$ ,  $\text{C=O}$ ) linked to glucuronic acid units of xylan, and the  $\text{CO}_2$  peak reported high temperatures are associated with thermal decomposition of cellulose. However, extrapolating these studies to the composition of bio-oil, the formation of carbon dioxide may be the result of the decarboxylation reactions of sugars present in the bio-oil. However, in the case of oxidation, carbon dioxide (Fig. 12f) is mainly the product of the oxidation of high molecular weight compounds remaining after the first stages of evaporation of light compounds.

#### 4. Conclusions

This investigation presents a TG-FTIR study on the pyrolysis and oxidation of bio-oil from sugarcane bagasse, using three heating rates: 5, 10, and 20 °C/min, and the compositional analysis of the evolved gases. From the TG/DTG oxidation curves, three macroscopic stages occurring during the thermal degradation of SCB bio-oil were identified. The low-temperature oxidation (LTO) stage was characterized by the release of water and oxygenates as identified from the FTIR spectra, for 20 °C/min it occurs at temperatures below 350 °C. Fuel decomposition (FD) stage identified by DTG analysis and kinetic analysis, identified between 350 and 450 °C and high-temperature oxidation (HTO), which corresponds to a clear signal in FTIR of oxidation products such as  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ , and  $\text{CO}$ , and occurs at temperatures above 450 for a heating rate of 20 °C/min.

In the case of pyrolysis, after the devolatilization stage, the formation of char occurs, the fixed carbon obtained was 11.5% of the initial mass and is a result of the decomposition of pyro-lignin which is the heaviest fraction of bio-oil. In the formation of char it is possible to observe an FTIR signal assigned to  $\text{CO}_2$  and some oxygenated compounds released as a result of the re-polymerization reactions. In the case of oxidation, although the signals of the aromatic groups released during devolatilization are very similar, the intensity of the  $\text{CO}_2$  peak is ten times higher, this because it is a product of oxidation at high temperatures.

From the kinetic analysis it was concluded that the thermal decomposition of the compounds present in the bio-oil leads to the formation of high molecular weight organic structures (char). These organic structures interact with the oxygen present in the air and this is absorbed at the surface as a previous step to oxidation at high temperatures. The Miura and Maki method is suitable for this analysis, since it allowed determining that the activation energy during the devolatilization stage for a heating rate of 5 °C/min varies between 50 and 100 kJ/mol. The decomposition stage of the fuel has a maximum activation energy value to be reached of 176 kJ/mol, and in the case of high temperatures oxidation the maximum energy that must be reached to trigger the oxidation is 320 kJ/mol.

#### CRediT authorship contribution statement

**Javier Ordonez-Loza:** Conceptualization, Methodology, Writing – original draft, Software, Validation. **Farid Chejne:** Conceptualization, Supervision. **Abdul Gani Abdul Jameel:** Conceptualization, Writing – review & editing. **Selvedin Telalovic:** Investigation. **Andrés Amell Arrieta:** ESI-FT-ICR experiments, Data discussion. **S. Mani Sarathy:** Supervision.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgments

The authors wish to thank the Colciencias-Doctorados Nacionales 757–2016 fellowship and the project "Strategy of transformation of the Colombian energy sector in the horizon 2030" funded by call 788 of the Colciencias Scientific Ecosystem. Contract number FP44842-210-2018. This work was performed at King Abdullah University of Science and Technology (KAUST) Clean Combustion Research Center (CCRC) with funding from the KAUST Center Applied Research Fund (CARF). This research used resources of the KAUST Catalysis Center (KCC).

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